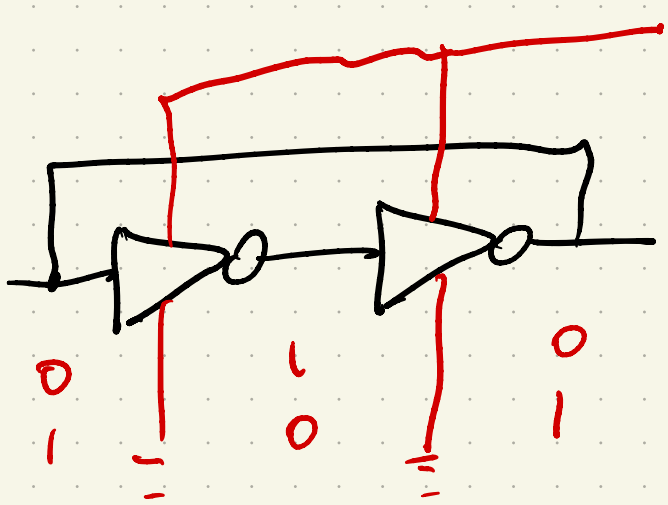
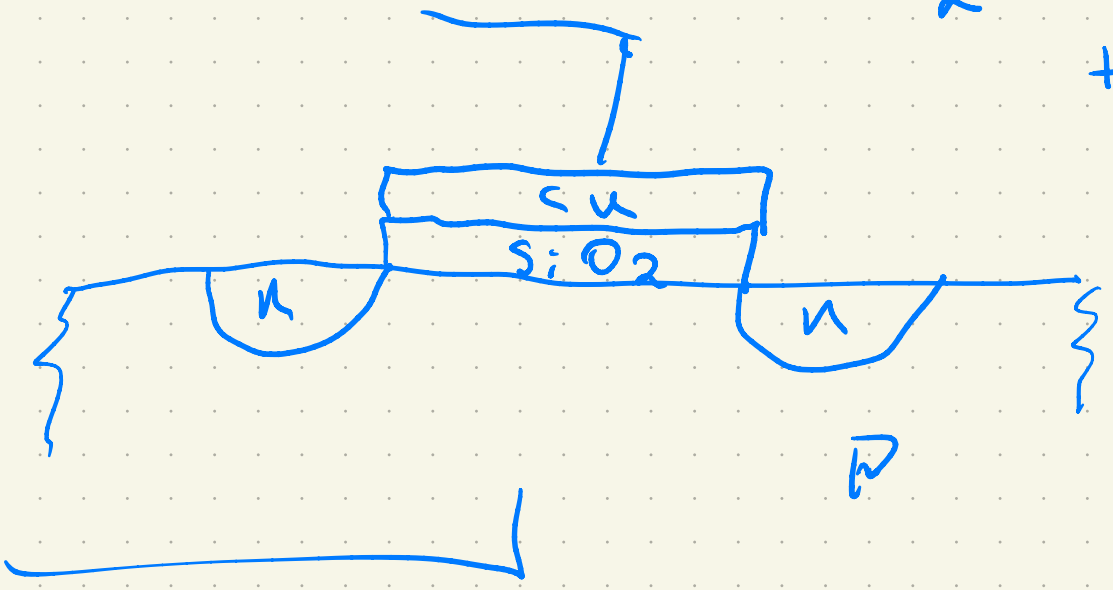
# Memory Technologies:

## SRAM




6 transistors

2 + 2 transistors
+ 2 addressing




Capacitance

(build memory out of any tech. that delays its input)

Cap. memory 2 transistors
1 capacitor
1 addressing

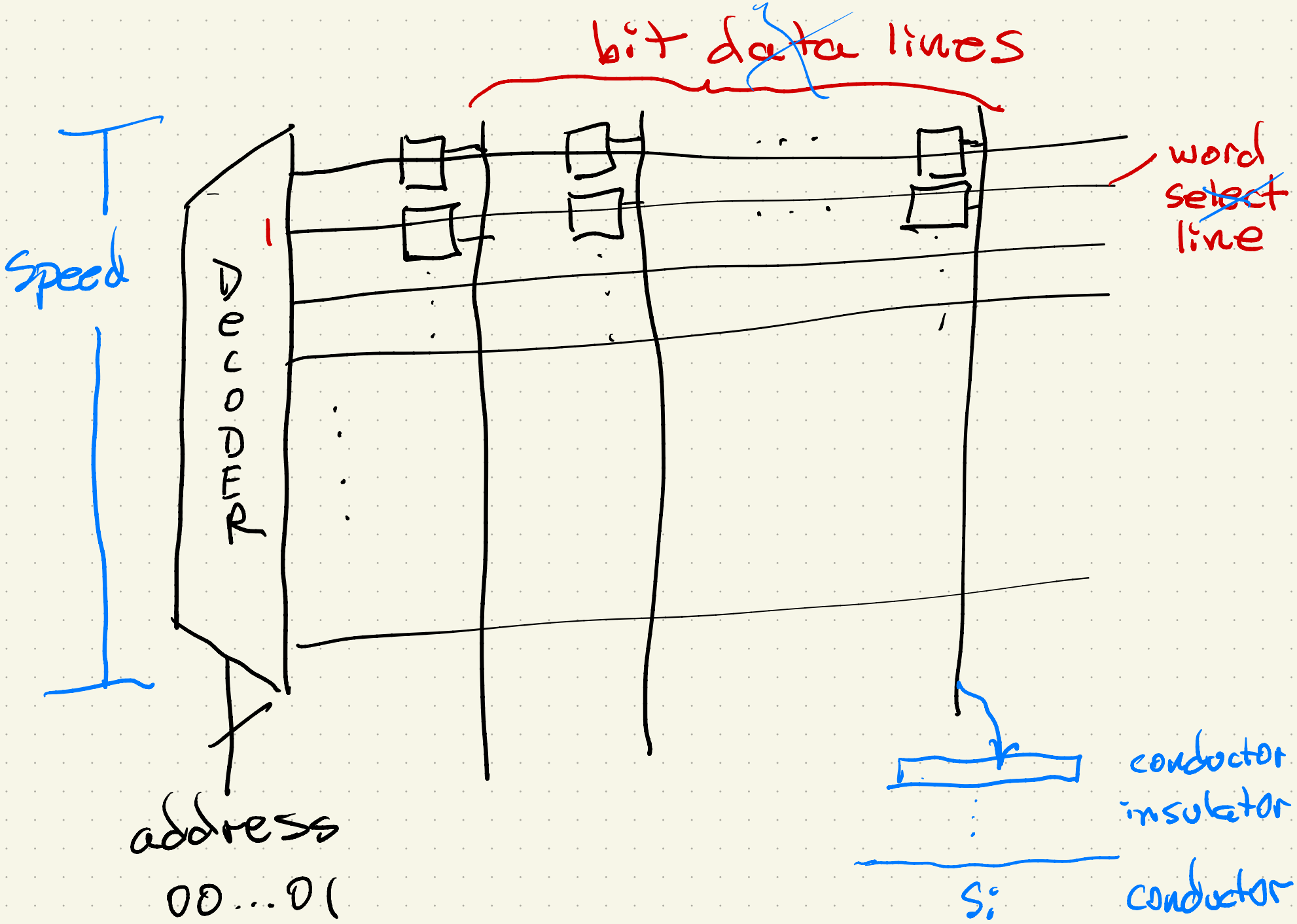
Denser than the SRAM way

Memory ~~that~~ forgets

DYNAMIC memory

DRAM

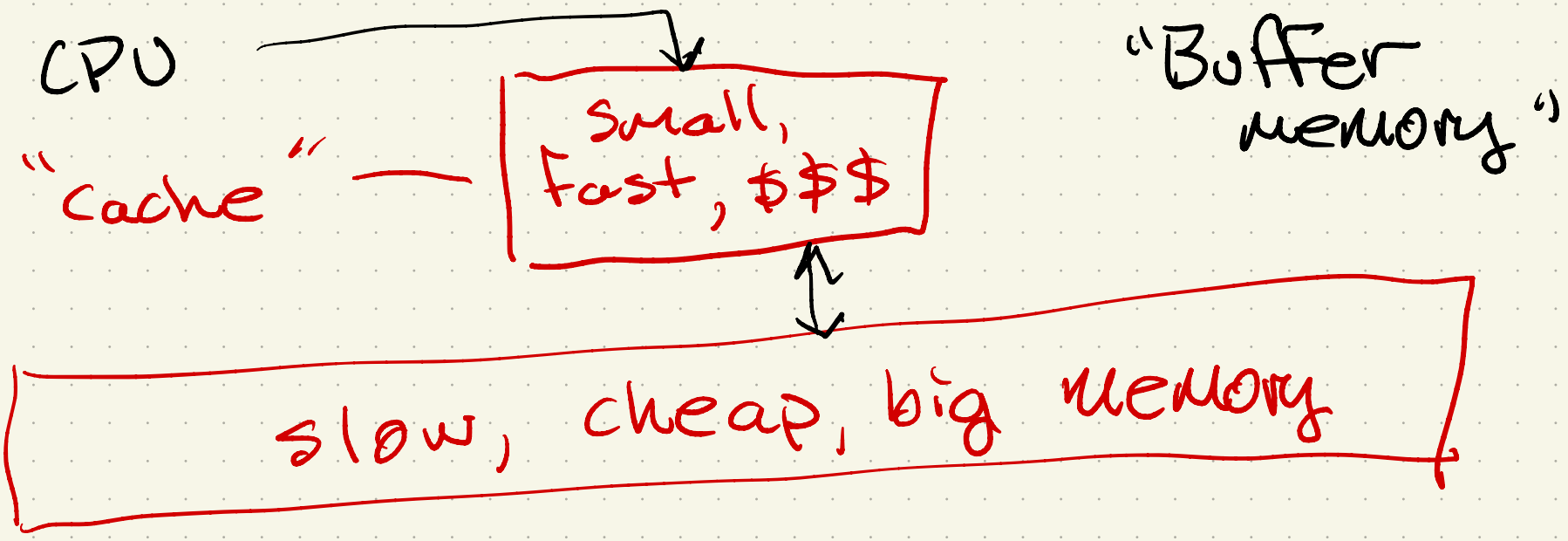
Destructive, slow, cheap


bit data lines
word select line
Speed
DECODER
1
address
00...01
conductor
insulator
conductor
Si

# Truth about memory

Either get Fast memory
or cheap memory,
but not both

Want Fast & cheap!

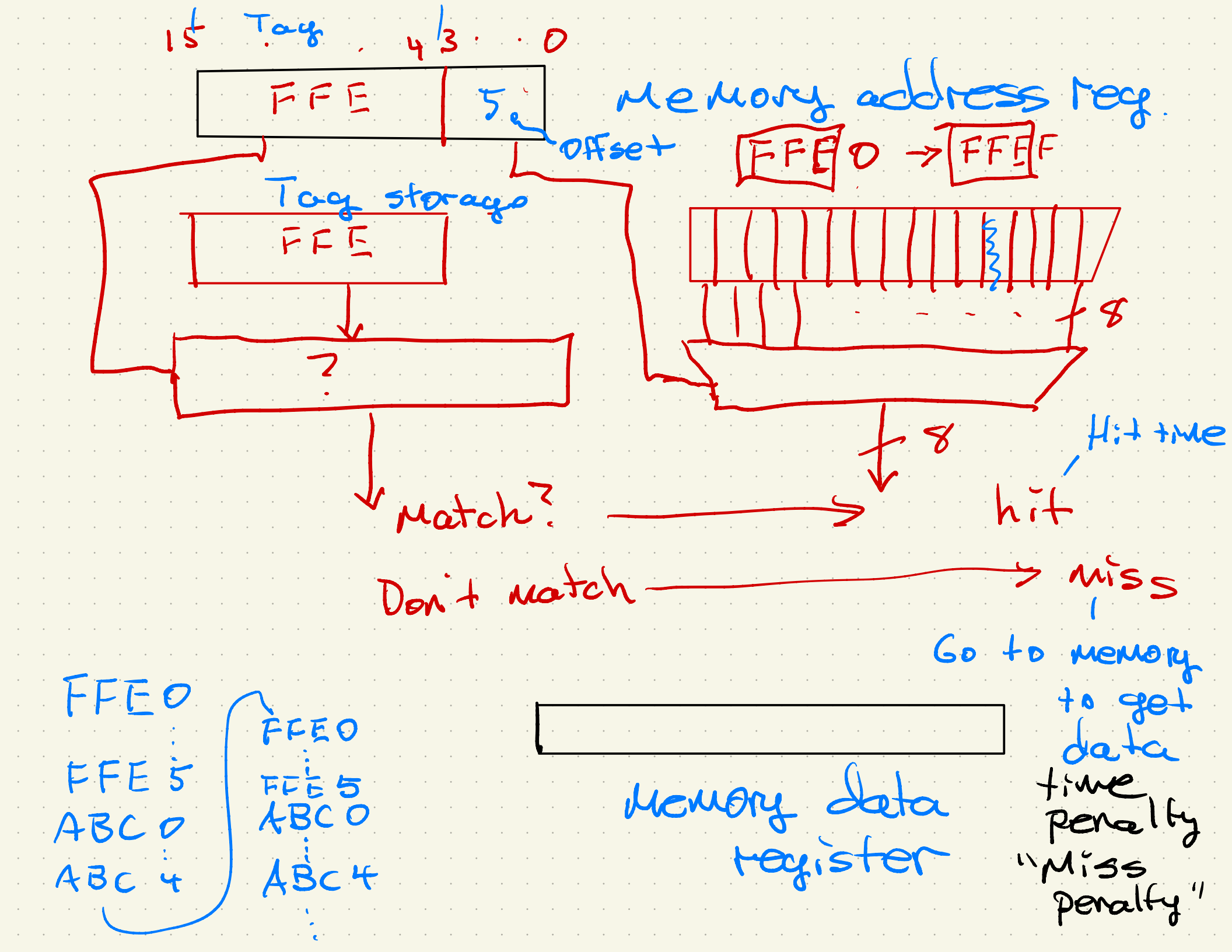


# Speculation + Prediction

Known program behavior

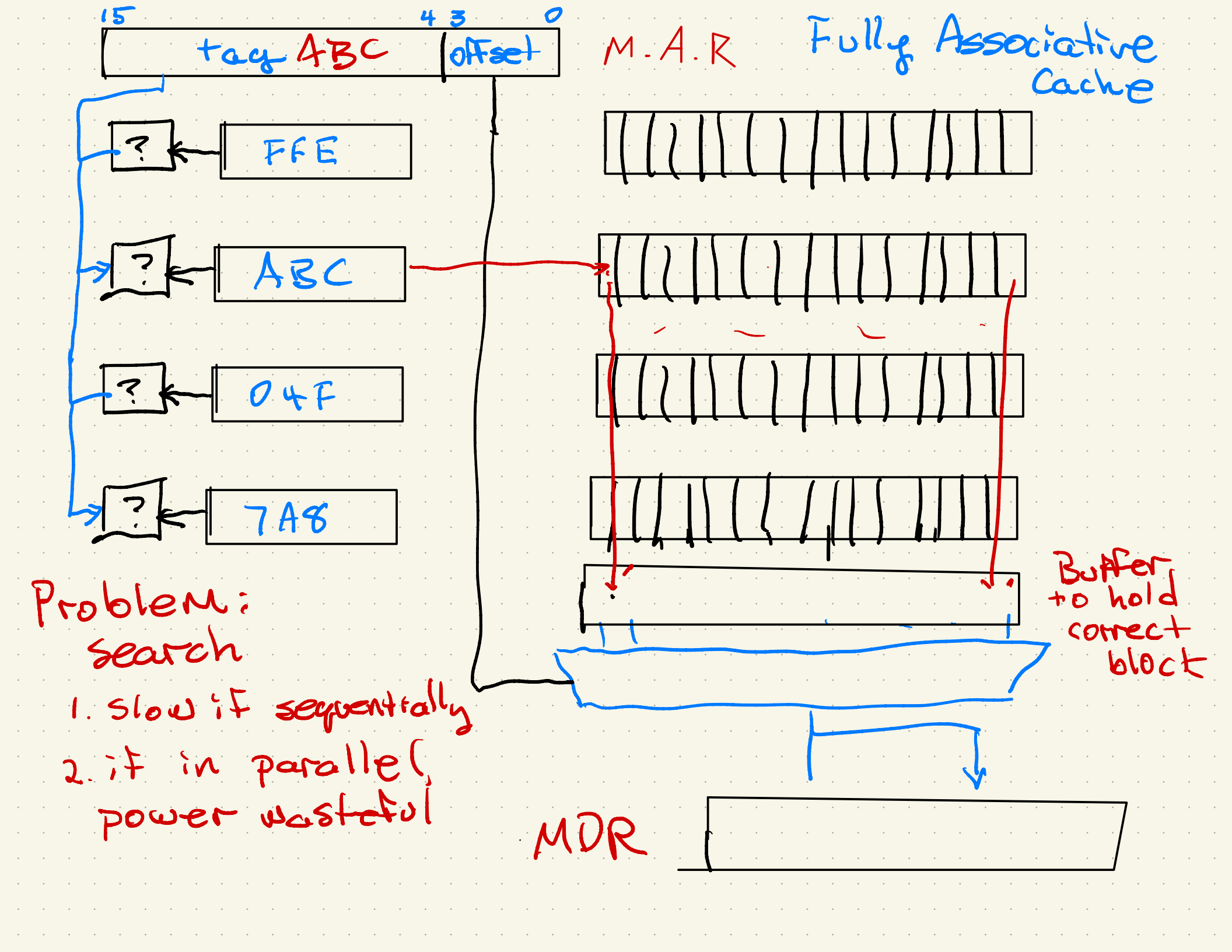
① Spatial locality
if access memory at address X at time t, highly likely to address $X \pm A$ at time $t \pm D$ for small A, D

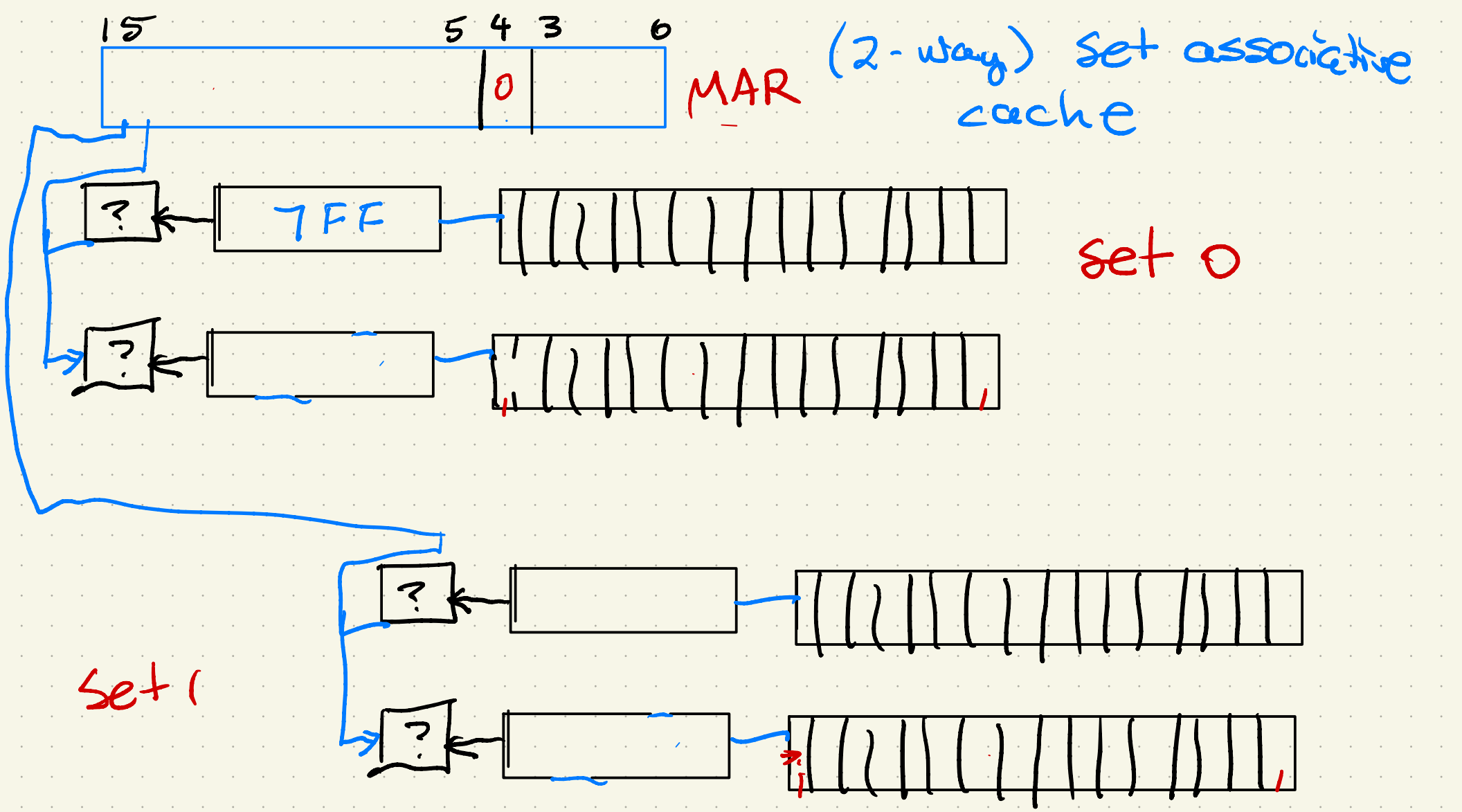
(1.5) Sequential locality
$A$ = constant $B$
$D = 1$
if access memory at X at t, highly likely to address $X + B$ at $t+1$ for constant B


15 Tag 4 3 0
FFE
5
offset
Memory address reg.
FFE0 → FFEF
Tag storage
FFE
?
8
8
Hit time
Match?
hit
Don't match
miss
Go to memory to get data
time penalty
"Miss penalty"
FFE0
FFE5
ABC0
ABC4
FFE0
FFE5
ABC0
ABC4
Memory data register

## (2) Temporal locality

IF access X at time t,
then highly likely to acess
X at time $t \pm D$ for small D


15
4 3
0
tag ABC
offset
M.A.R
Fully Associative Cache
?
FFE
?
ABC
?
04F
?
7A8
Problem:
search
1. slow if sequentially
2. if in parallel, power wasteful
Buffer to hold correct block
MDR


15
5 4 3
0
6
MAR
(2-way) set associative cache
?
7FF
set 0
?
?
set 1
?